THE CHALLENGE Police officers in the field frequently find themselves in danger of an incoming attack just because they don't have eyes in the back of their heads.
The challenge is to create a "Virtual Trustzone" around police officers to detect or predicts an incoming attack from the surrounding area and alert officers in time for effective response.

THE SOLUTION incorporates state of the art TensorFlow Deep Learning Algorithms for person detection and OpenCV software for motion detection. It will leverage MSI's Flex360 body worn camera which incorporate both front & rear facing camera.
TensorFlow is Google's open-source software library used in many wide range solutions. Our solution incorporates TensorFlow with a camera for fast detection of an person in the camera's field of view. TensorFlow average detection time is about 500 msec which translated into detecting person running 15 km/h in approximately 2 meters from 10 to 15 meters.
OpenCV is an open source vision software library with, motion detection algorithm, which allow detection of an approaching object.

OUR NEXT STEP To enhance performance and create a robust solution we plan to:

- Replace TensorFlow with better person detection solution, such as Neurala that is specially designed for person search
- Incorporate Neurala with motion sensors to detect fast approaching object

PAVEL LIFSHITZ

DAVID LEV

ALEX RIVKIN

YOTAM KESHET

ALEX APARSTIN